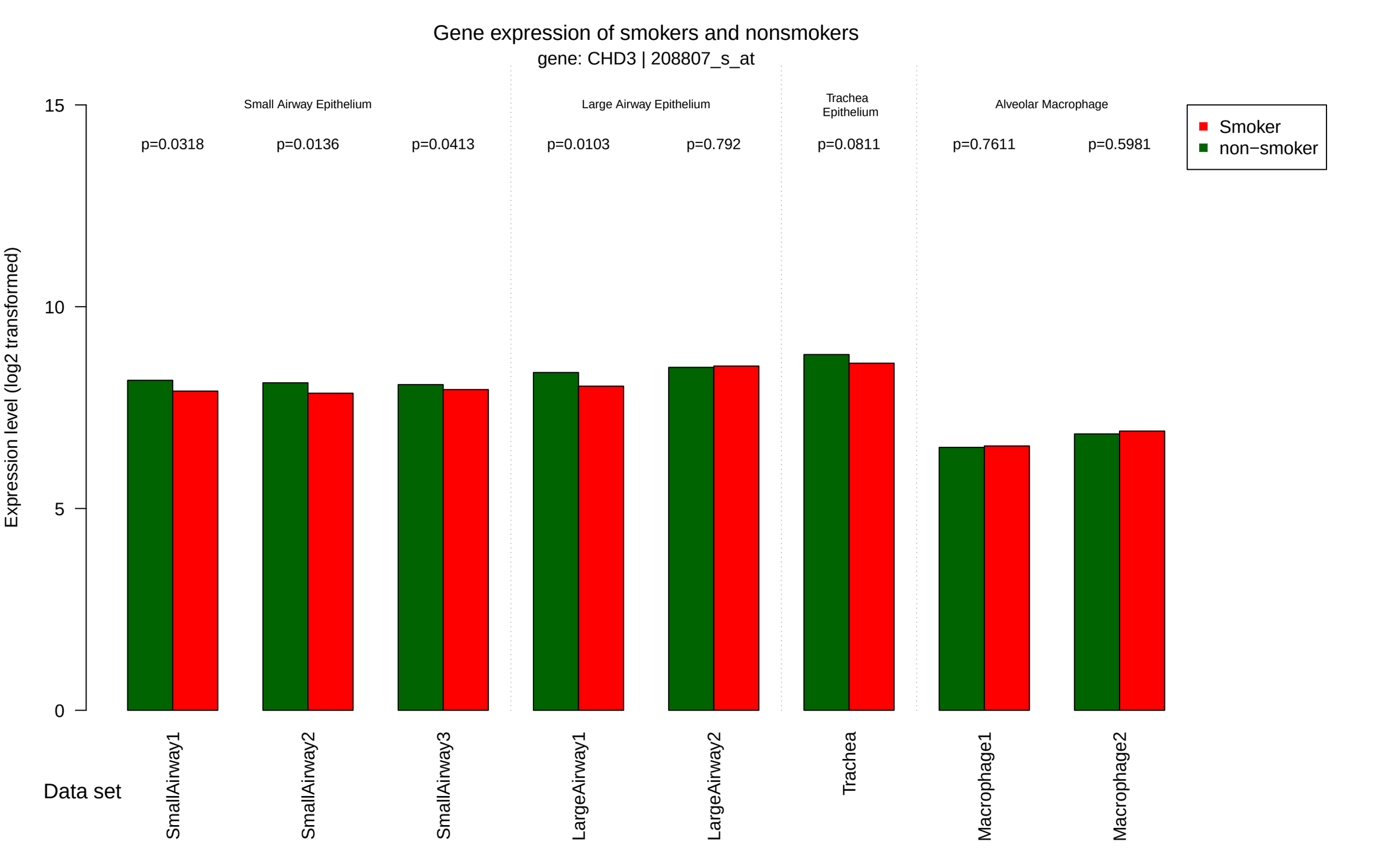
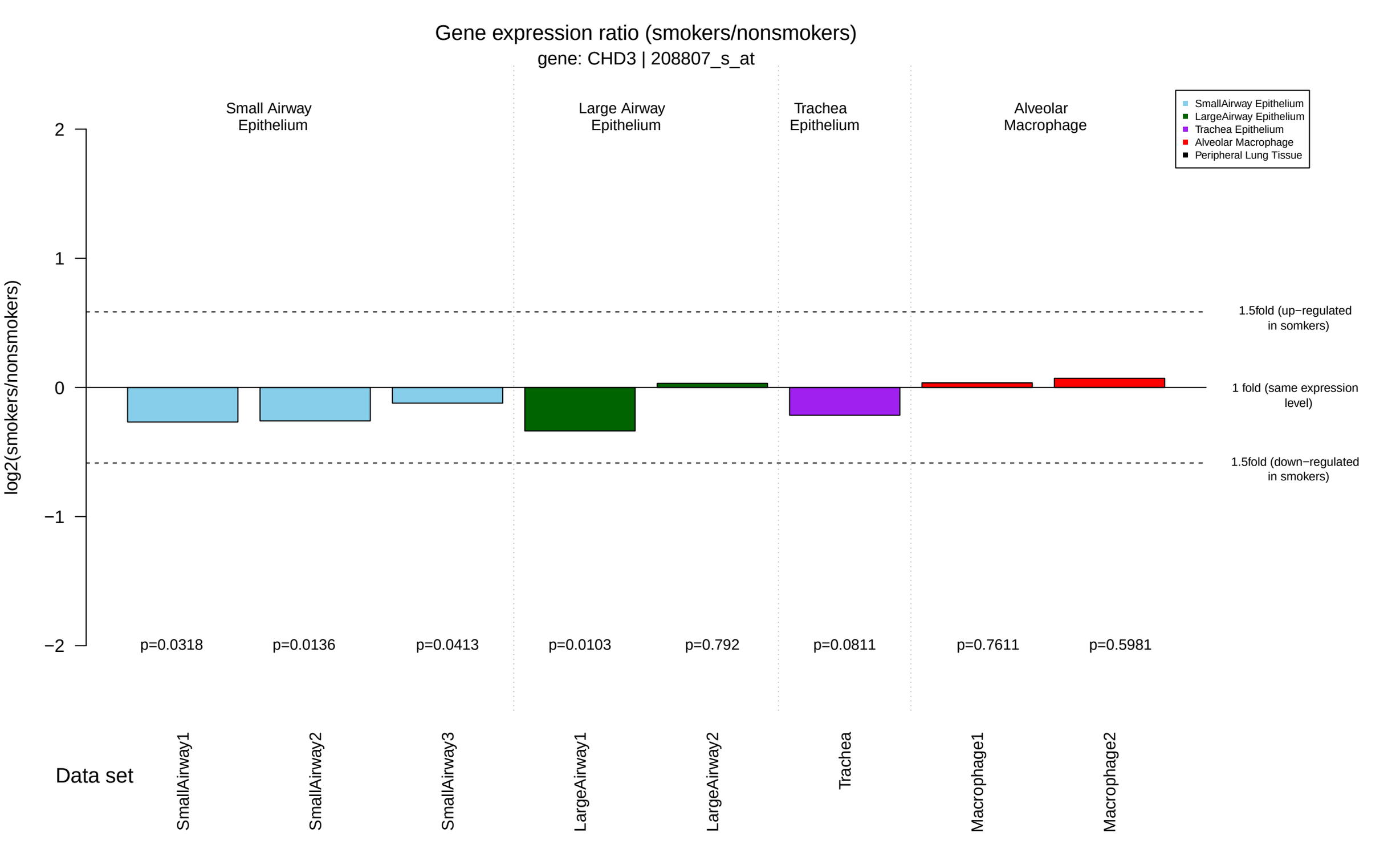
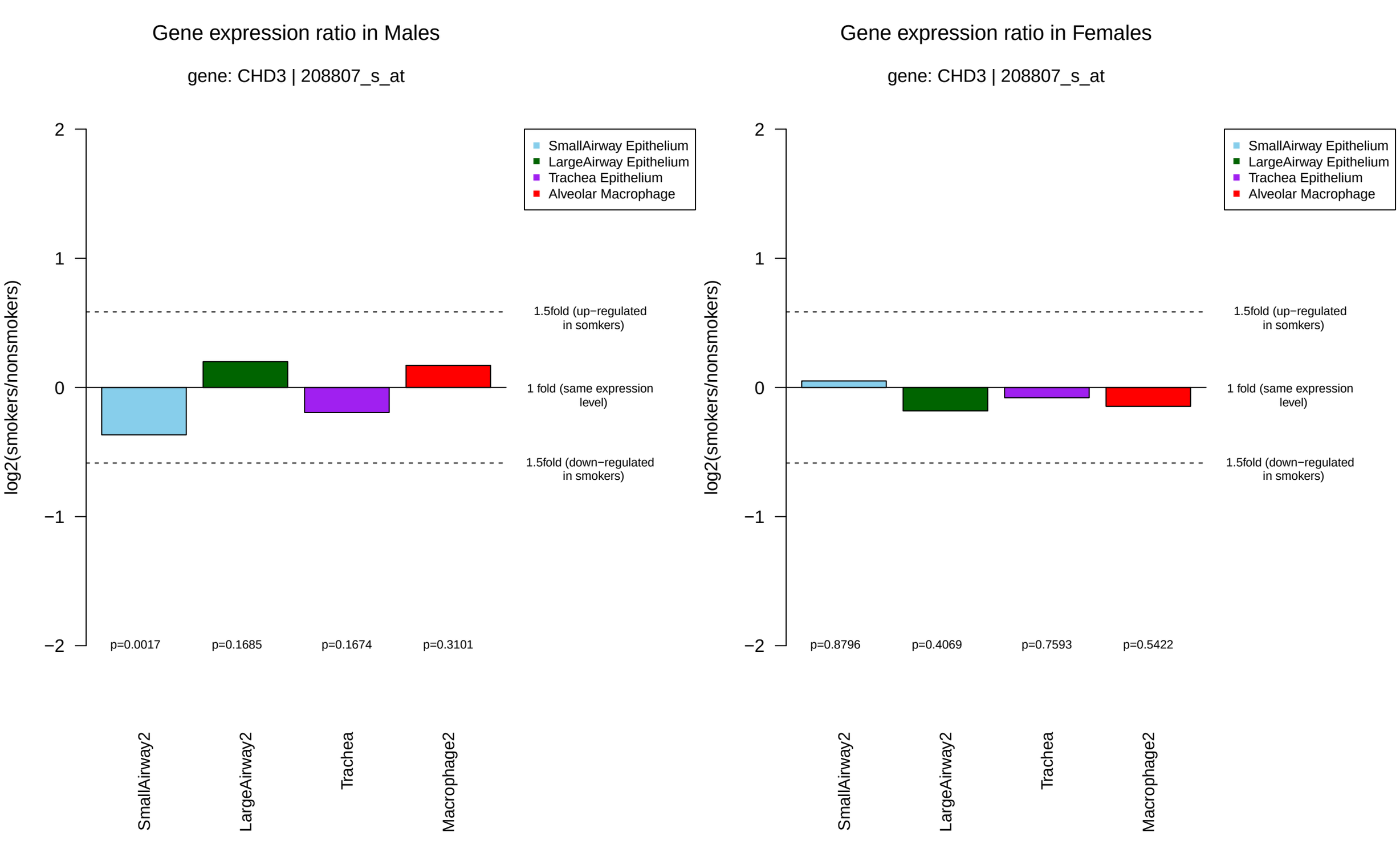
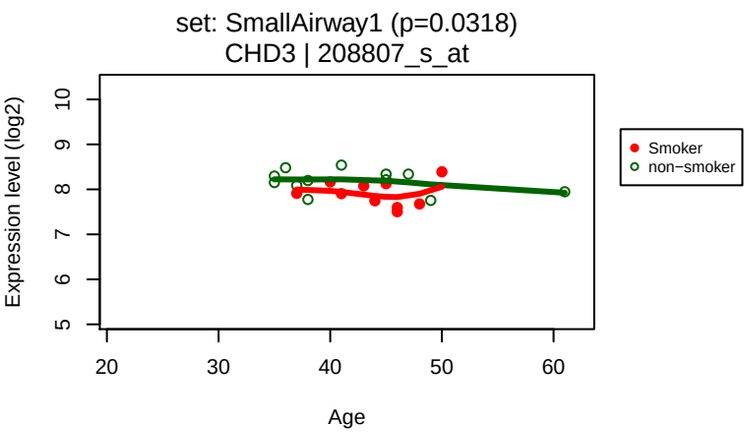
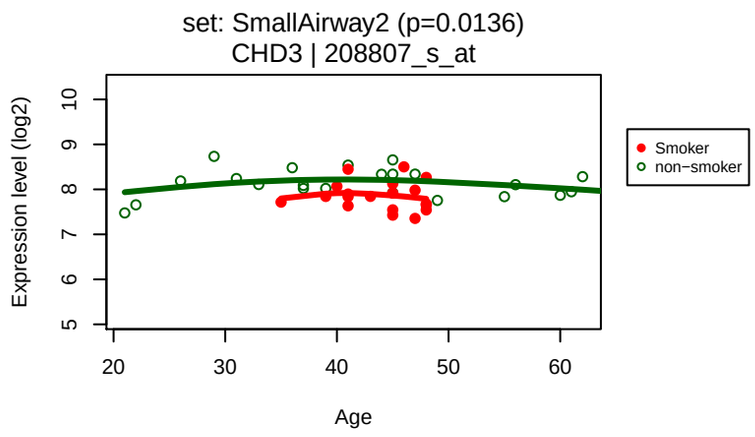
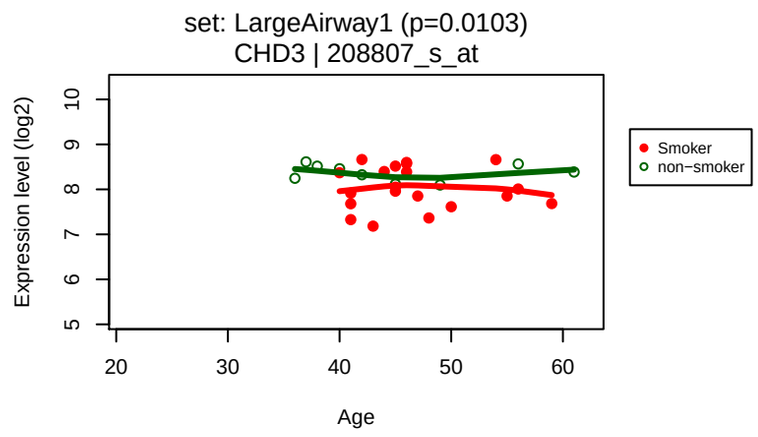

Gene expression of smokers and nonsmokers
gene: CHD3 | 208807_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0318
p=0.0136
p=0.0413
p=0.0103
p=0.792
p=0.0811
p=0.7611
p=0.5981
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

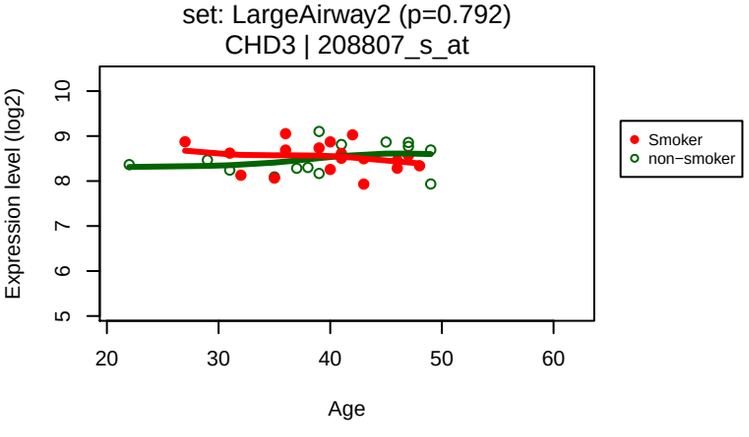

Gene expression ratio (smokers/nonsmokers)
gene: CHD3 | 208807_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0318
p=0.0136
p=0.0413
p=0.0103
p=0.792
p=0.0811
p=0.7611
p=0.5981
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

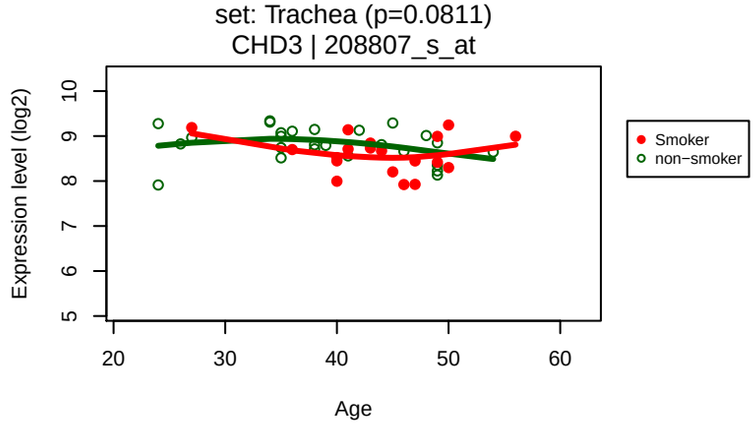

Gene expression ratio in Males
gene: CHD3 | 208807_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.1685
p=0.1674
p=0.3101
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD3 | 208807_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8796
p=0.4069
p=0.7593
p=0.5422
SmallAirway2
LargeAirway2
Trachea
Macrophage2

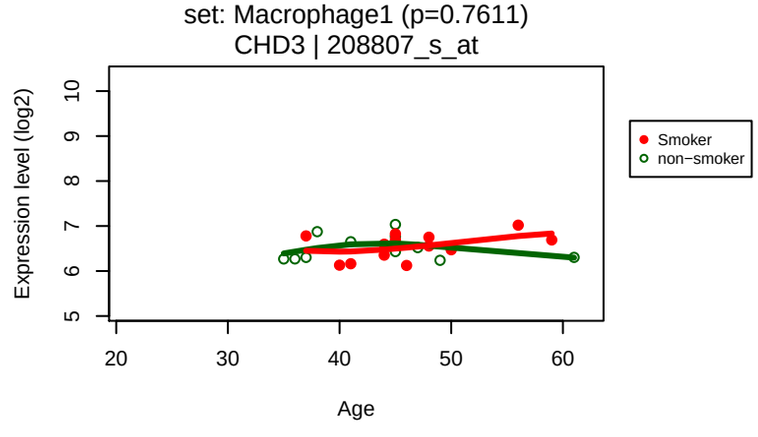

set: SmallAirway1 (p=0.0318)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker

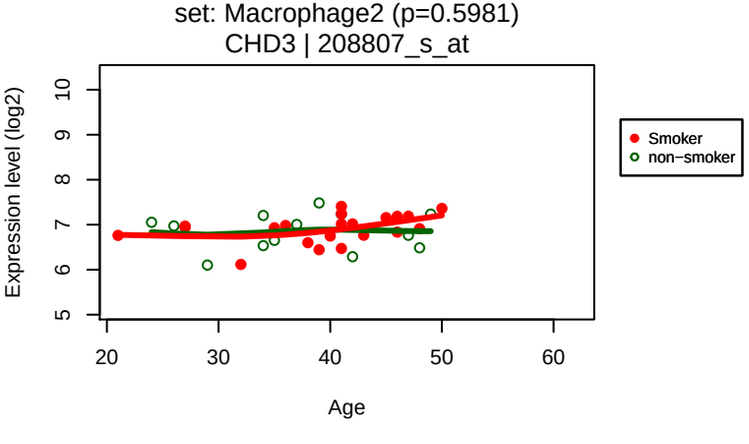

set: SmallAirway2 (p=0.0136)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker

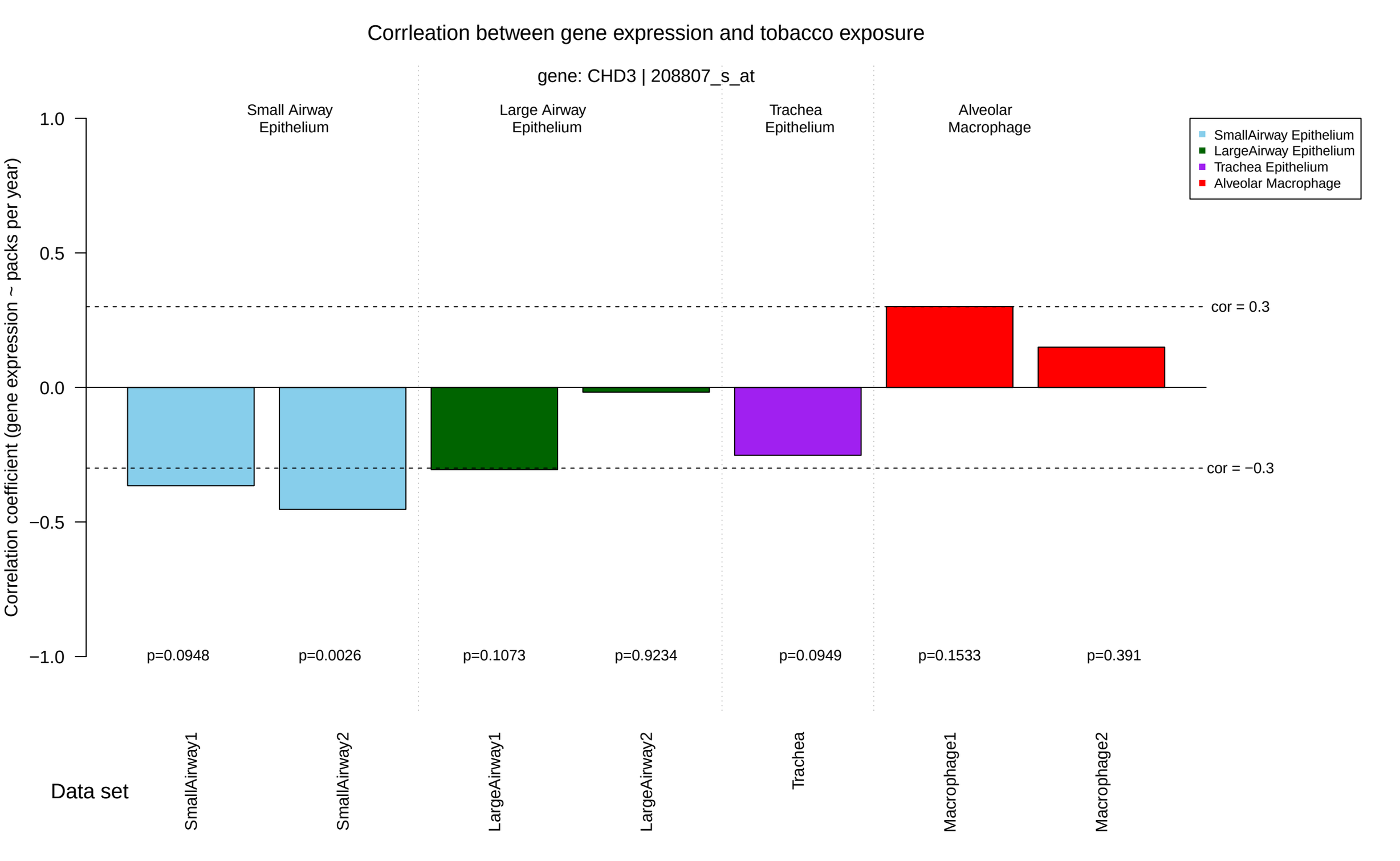

set: LargeAirway1 (p=0.0103)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.792)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0811)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7611)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5981)
CHD3 | 208807_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD3 | 208807_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0948
p=0.0026
p=0.1073
p=0.9234
p=0.0949
p=0.1533
p=0.391
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2